

Kian Javaheri

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EDUCATION

Arizona State University

Aug. 2022 – May 2026

Bachelor of Science in Computer Science, Bachelor of Science in Economics

Tempe, AZ

- **GPA: 3.93/4.00** | Barrett, The Honors College

Relevant Coursework: Discrete Math · Data Structures & Algorithms · Operating Systems · Software Analysis & Design · Intro to Artificial Intelligence

EXPERIENCE

R&D Bioresource and Environment Security Intern

May 2023 – Aug. 2024

Sandia National Laboratories

Livermore, CA

- Engineered a **standalone optimization tool** that digitizes decision-making across multiple physical bioresource systems, ingesting real-world operational inputs and **maximizing profit yields** using configurable **optimization algorithms**, reducing manual analysis time for research teams
- Developed a **QGIS geospatial plugin** that parses **spatial datasets** and turns raw geographic data into optimal feedstock sourcing decisions based on user-defined material quantities and cost constraints

Undergraduate Research Assistant, Dr. Michael Hanemann

May 2025 – May 2026

Arizona State University

Tempe, AZ

- Automated large-scale **document retrieval** from the Arizona Department of Water Resources using **Selenium WebDriver**, eliminating manual collection across hundreds of annual reports
- Built **OCR extraction pipelines** to parse and structure tabular data from scanned PDFs, producing clean, analysis-ready datasets for downstream **econometric modeling**

PROJECTS

Revolution Parts Shipping Service with Caching | *React, PostgreSQL, Flask, Docker* Aug. 2025 – May 2026

- Architected a **high-throughput relational caching layer** using **PostgreSQL** and **Flask** within a **Dockerized** microservice environment, intercepting and caching external carrier **API** responses to reduce redundant network calls
- Achieved a **64.4% reduction in processing latency** (modal response time: 334ms → 119ms), delivering a **2.8x speedup** over live API calls and substantially reducing data-layer overhead

Materials GUI | *Java, Fabric, Claude API*

Current

- Built a client-side **Fabric mod** for Minecraft that tracks the materials needed for a build, letting players import a material list by pasting text, loading a Litematica export, or dropping in a screenshot parsed via **OCR** through the **Claude API**
- Implemented real-time inventory tracking across in-game storage containers, color-coding each slot by completion progress and auto-routing shift-clicked items to the slots still needing materials

LEADERSHIP & ACTIVITIES

Undergraduate Teaching Assistant

Aug. 2023 – May 2024

Arizona State University

Tempe, AZ

CSE310: Data Structures & Algorithms

- Led exam review sessions on **red-black trees**, **dynamic programming**, and **minimum spanning tree** algorithms, and held 5+ office hours per week guiding students through **algorithm design** challenges

Google Data Analytics Certificate

Oct. 2022

Coursera

- Completed all 8 courses covering **data cleaning**, **SQL querying**, **data visualization**, and **statistical analysis** using R and Tableau

TECHNICAL SKILLS

Machine Learning & AI: Fine-Tuning, Transformers, PyTorch, PySpark, Feature Engineering, Predictive Modeling

Languages: Java, Python, C/C++, SQL, JavaScript/TypeScript, HTML/CSS, R

Frameworks: React, Node.js, Flask, JUnit

Developer Tools: Git, GitHub, Docker, JupyterHub, Firebase, MATLAB, QGIS, Postman, Stata, Qualtrics

Libraries: Pandas, NumPy, Matplotlib, Selenium